

深圳市金逸晨电子有限公司

ShenZhen GoldenMorning Electronic Co.,Ltd

PRODUCT SPECIFICATION

Model No	T180BV-C20-02
Product Name	1.8 TFT LCD Module
Version	V1.0
Date	2025-11-20

- ☐ Preliminary Specification
☒ Final Specification

PREPARED BY	CHECKED BY	APPROVED BY

Revision History

Version	Date	Page	Revise record	Remarks
V1	2025.11.20	-	First issued.	ZY

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1. General Parameter

NO.	Features	Details	Unit
1	Display Size(Diagonal)	1.8"	Inch
2	LCD type	TFT	-
3	Display Mode	Normally black	-
4	Resolution	360(H)RGB x360(V)	Pixels
5	Viewing Direction	All View	Best image
6	Module size	47.99(H) x 49.86(V) x1.92(D)	mm
7	Active Area	Ø45.68	mm
8	Dot pitch	0.1269(H) x 0.1269(V)	mm
9	Color arrangement	RGB Vertical stripe	-
10	Interface	QSPI	-
11	Display Colors	16.7M	-
12	Driver IC	ST77916	-
13	Backlight	3 White LED Parallel	-
14	Operating Temperature	-20℃~70℃	℃
15	Storage Temperature	-30℃~80℃	℃
16	Weight	-	g

Note 1: Please refer to the mechanical drawing.

2. Physical drawing



[illegible]

4. Absolute Maximum Ratings

Item	Symbol	Min.	Max.	Unit
Power Supply for Analog	VDD	-0.3	4.6	V
Power Supply for I/O System	VDDI	-0.3	4.6	V
Operating temperature	Top	-20	70	°C
Storage temperature	Tst	-30	80	°C

Note 1: If Ambient temperature below 50°C, the maximal humidity is 90%RH, if Ambient temperature over 50°C, absolute humidity should be less than 60%RH.

Note 2: The response time will be extremely slow when the operating temperature is around -10°C, and the back ground will become darker at high temperature operating.

Note 3: If one of the above items is exceeded its maximum limitation momentarily, the quality of the product may be degraded. Absolute maximum limitation, therefore, specify the values exceeding which the product may be physically damaged. Be sure to use the product within the recommend range.

5. DC Characteristics

Parameter	Symbol	Min	TYP	MAX	Unit	Notes
Power Supply for Analog	VDD	2.65	2.8	3.3	V	
Power Supply for I/O System	VDDI	1.65	1.8	3.3		
Gate Driver High Voltage	VGH	11	-	15.5	V	
Gate Driver Low Voltage	VGL	-11.7	-	-8.4	V	
Operating Current for V _{DD}	I _{DD}	-	8	10	mA	
Sleep_In Mode VDD	I _{dd}	-	30	150	uA	
Sleep_In Mode VDDIO	I _{ddio}	-	1	5	uA	

6. Backlight Characteristic

Parameter	Symbol	Min	TYP	MAX	Unit	Notes
Voltage for LED backlight	VLED	2.8	2.9	3.0	V	
Current for LED backlight	ILED	-	60	-	mA	3 LED
Power Consumption	Pbl	-	174	-	mW	1
Brightness	L _{br}	550	600	-	cd/m ²	2
LED Life time	-	20000	-	-	hr	3

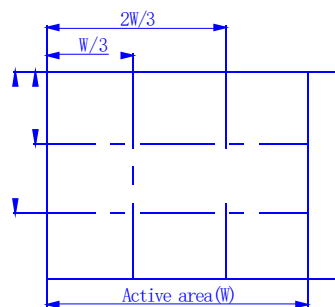
1.Where ILED =60mA , VLED=3.0V , Pbl= ILED x VLED

2.Uniform measure condition:

a:Measure 9 point ,Measure location is show below:

b:Uniform=(Min brightness/Max brightness)x100%

c:Best Contrast.

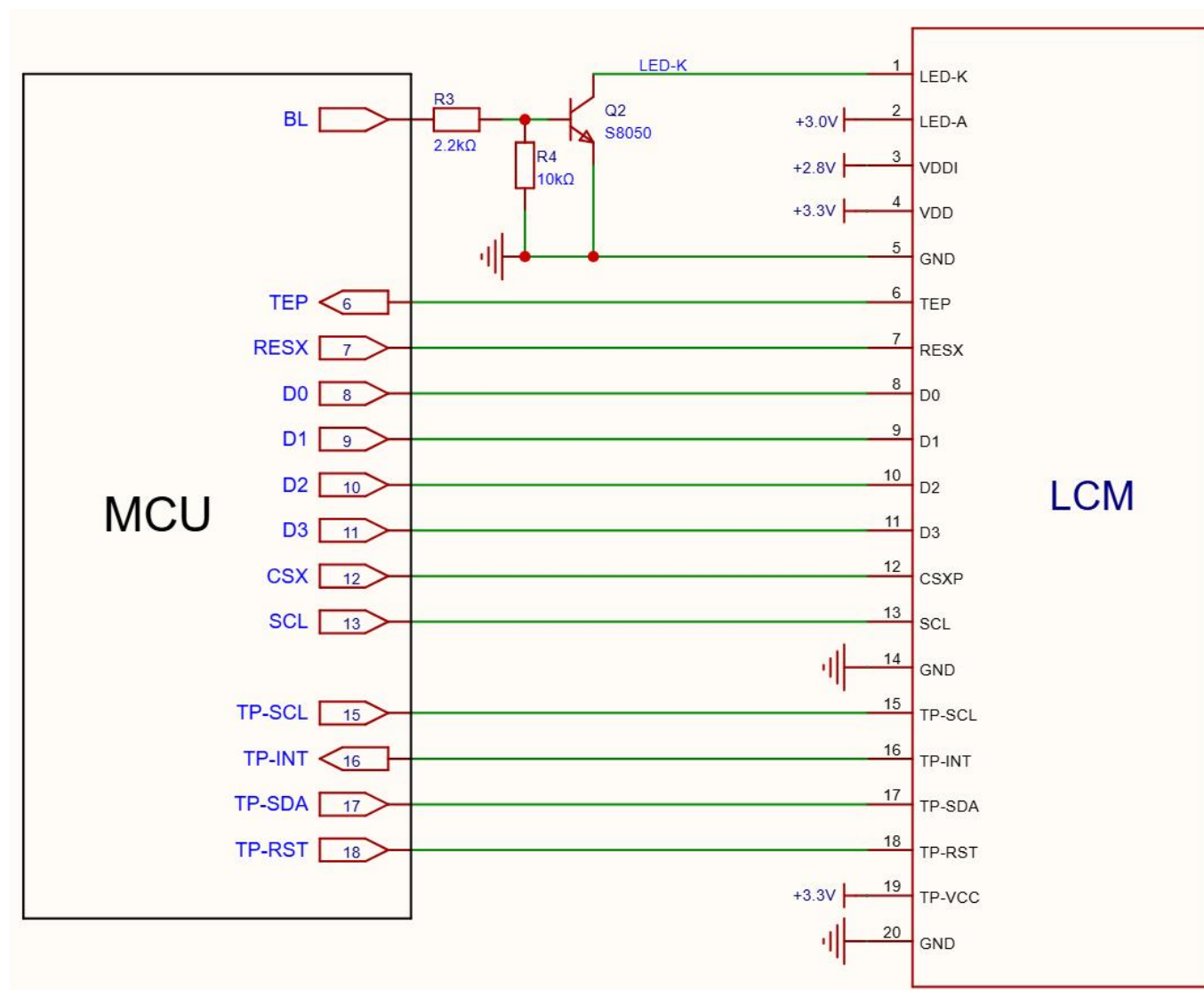


3.The environmental conducted under ambient air flow ,at Ta=25±2oC,60%RH±5%

7. Pins Definitions.

NO.	Symbol	Description
1	LED-K	LED Cathode
2	LED-A	LED Anode
3	VDDI	Power Supply for I/O System
4	VDD	Power Supply for Analog, Digital System and Booster Circuit.
5	GND	Power Ground.
6	TEP	Tearing effect signal is used to synchronize MCU to frame memory writing.
7	RESX	This signal will reset the device and it must be applied to properly initialize the chip .Signal is active low.
8	D0	Serial input signal in QSPI serial data interface
9	D1	Serial input signal in QSPI serial data interface
10	D2	Serial input signal in QSPI serial data interface
11	D3	Serial input signal in QSPI serial data interface
12	CSXP	Chip selection pin ;Low enable ,high disable.
13	SCL	This pin is used to be serial interface clock
14	GND	Power Ground.
15	TP-SCL	Touch Serial clock pin.
16	TP-INT	Touch interrupt input pin
17	TP-SDA	Touch Serial data input / output.
18	TP-RST	Touch Reset, low active, weak pull up
19	TP-VCC	Touch Power supply for Analog
20	GND	Power Ground.

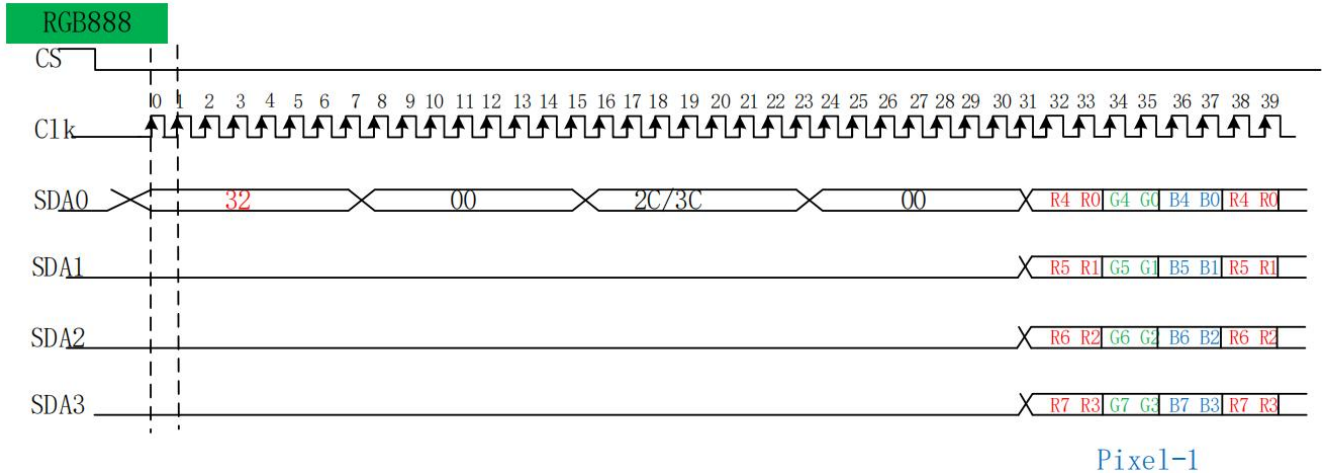
8. Schematic Diagram



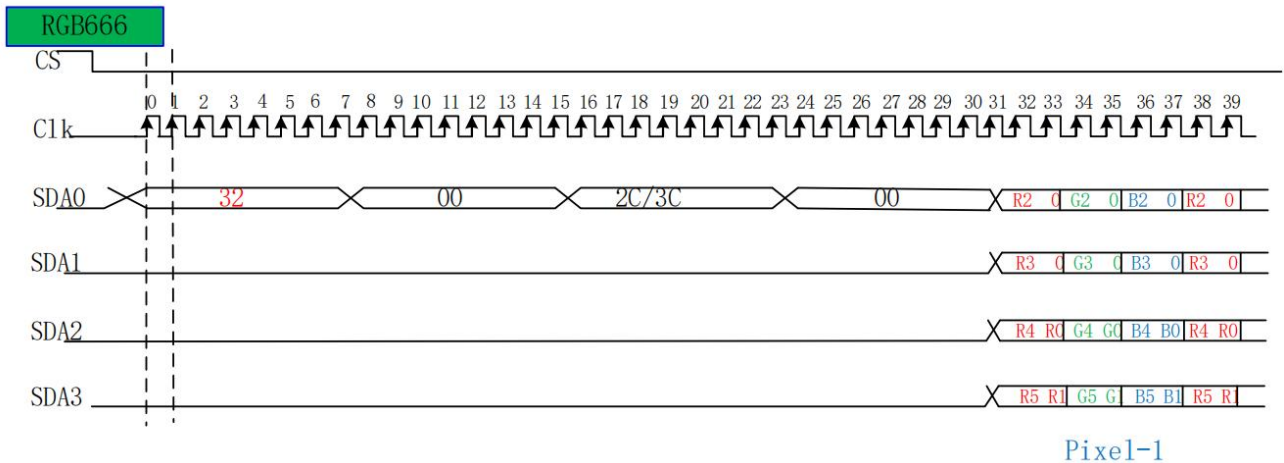
9. Timing Characteristics

4 wire data 1 wire Addr: first byte=0x32

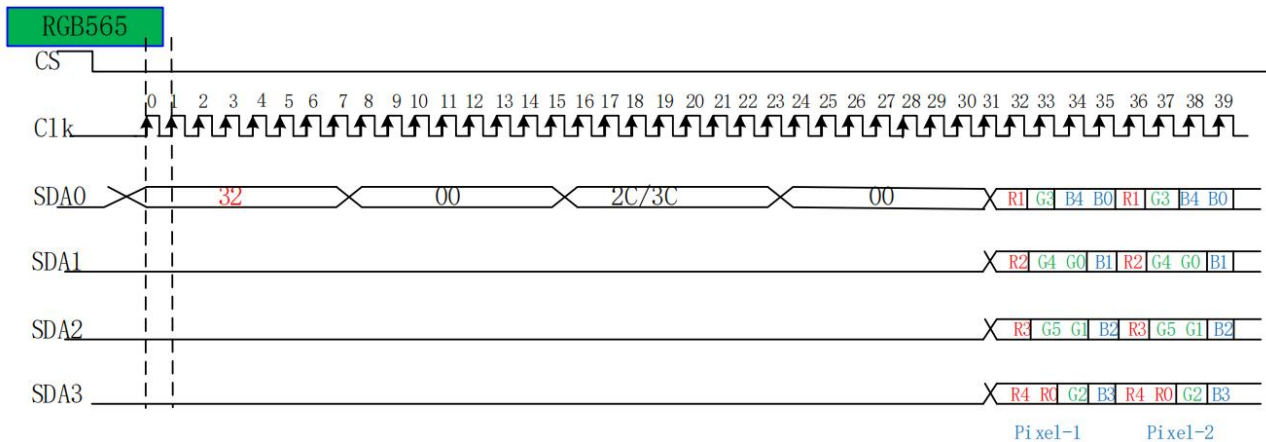
1) 16.7M-Colors:24-bit/pixel (RGB 8, 8, 8 -bits input).



2) 262k-Colors:18-bit/pixel (RGB 6, 6, 6 -bits input).



3) 65k-Colors:16-bit/pixel (RGB 5, 6,5 -bits input).

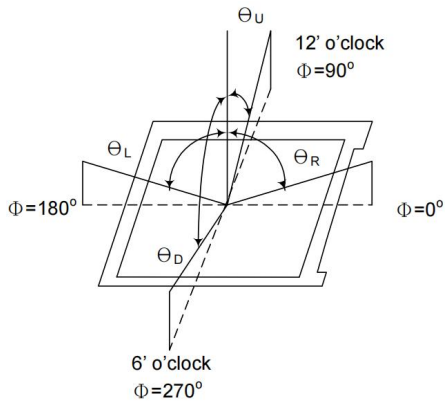


Please refer to the driver IC specification

10. Optical Characteristics

Item	Symbol	Measuring Conditions		Min.	Typ.	Max.	Unit	Remark
Viewing Angle		= 0°	25 °C	80	85	-	Deg	Note1
		=180°	25 °C	80	85	-		
		= 90°	25 °C	80	85	-		
		=270°	25 °C	80	85	-		
Brightness	Lbr	--	-	550	600	-	Cd/m2	
Luminance Uniformity	ΔL	--	-	75	80	-		
Contrast Ratio	CR	--	25 °C	1000	1200	-	--	Note2
Response Time	Tr+Tf	= 0° = 0°	25 °C	-	30	35	ms	Note3
Color of CIE Coordinate	White	X	25 °C	-0.03	0.278	+0.03	--	BM-7A
		Y	25 °C		0.297			
	Red	X	25 °C		0.672			
		Y	25 °C		0.304			
	Green	X	25 °C		0.284			
		Y	25 °C		0.666			
	Blue	X	25 °C		0.112			
		Y	25 °C		0.083			
Transmittance (with polarizer)	--	--	--	3.7	4.35	-	%	--

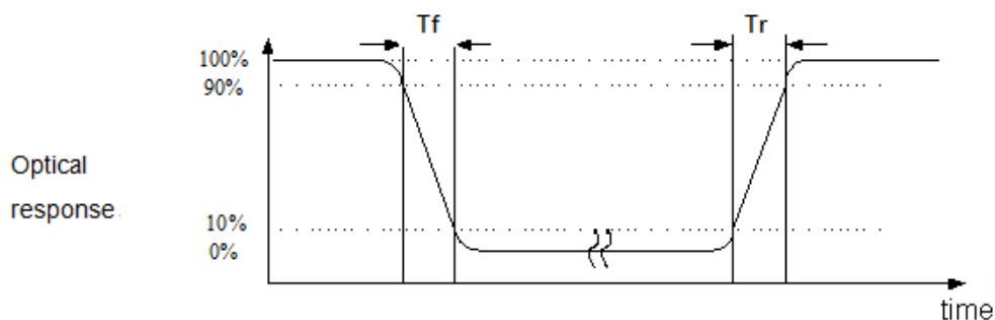
Note (1) Definition of Viewing Angle:



Note (2) Contrast ratio is calculated by the following formula:

$$\text{Contrast ratio (CR)} = \frac{\text{Brightness on the "white" state}}{\text{Brightness on the "black" state}}$$

Note (3) Definition of Response Time: Sum of TR and TF



11. Reliability

11.1. MTBF

The LCD module shall be designed to meet a minimum MTBF value of 50000 hours with normal.

11.2. Test Condition

No	Item	Condition	Criteria
1	High Temperature Operating	70℃*24Hrs	1. No Defect Of Operational Function In Room Temperature Are Allowable. 2. IDD of LCM in Pre-and Post-Test Should Follow Specification.
2	Low Temperature Operating	-20℃*24Hrs	
3	High Humidity Storage	50℃*90%RH*24Hrs	
4	High Temperature Storage	80℃*24Hrs	
5	Low Temperature Storage	-30℃*24Hrs	
6	Thermal Cycling Test Storage	-	
7	Packing vibration	-	
8	Electrical Static Discharge	-	
		-	
9	Drop Test (Packaged)	-	

Note1. The test samples should be applied to only one test item.

Note2. Sample size for each test item is 2pcs.

Note3. No defection function allowable.

12. Precautions

12.1. Storage Conditions

- 12.1.1. Store the panel or module in a dark place where the temperature is $23\pm5^{\circ}\text{C}$ and the humidity is below $45\pm 20\%\text{RH}$.
- 12.1.2. Store in anti-static electricity container.
- 12.1.3. Store in clean environment, free from dust, active gas, and solvent.
- 12.1.4. Do not place the module near organics solvents or corrosive gases.
- 12.1.5. Do not crush, shake, or jolt the module.
- 12.1.6. Strong light exposure causes degradation of polarizer and color filter.

12.2. Handling Precautions

- 12.2.1. Avoid static electricity, which can damage the CMOS LSI.
- 12.2.2. The polarizing plate of the display is very fragile, please handle it very carefully.
- 12.2.3. Do not give external shock.
- 12.2.4. Do not apply excessive force on the surface.
- 12.2.5. Do not wipe the polarizer with dry cloth, as it might cause scratch. If the surface of the LCD needs to be cleaned, wipe it swiftly with cotton or other soft cloth soaked with petroleum IPA, do not use other chemicals.
- 12.2.6. Reverse and use within ratings in order to keep performance and prevent damage
- 12.2.7. Do not remove the panel or frame from the module.
- 12.2.8. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320°C to 350°C , the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.

12.3. Limited Warranty

- 12.3.1. Our warranty liability is limited to repair and/or replacement. We will not be responsible for any consequential loss. (我们的保证责任仅限于修理和/或更换。我们将不负任何相应的损失。)
- 12.3.2. If possible, we suggest customer to use up all modules in six months. If the module storage time over twelve months, we suggest that recheck it before the module be used. (如果可能, 我们建议客户在 6 个月内用完所有模块。如果模块存放时间超过 12 个月, 我们建议在使用模块前重新检查。)
- 12.3.3. The warranty period is twelve months from the date of delivery. Buyer shall complete the assembly of all processes within twelve months of validity. During the warranty period, if the product quality problems, our company will be responsible for repair and replacement. All products must be stored and handled in accordance with regulations. Under warranty. When the goods do not comply with the above terms, we do not provide warranty services. (保修期为交货之日起十二个月。买方应在 12 个月内完成所有流程的组装。在保修期内, 如果产品出现质量问题, 我公司将负责维修和更换。所有产品必须按照规定储存和处理。在保修期内。当货物不符合上述条款时, 我们不提供保修服务。)